

Transformers

Module 2

DEFINITION

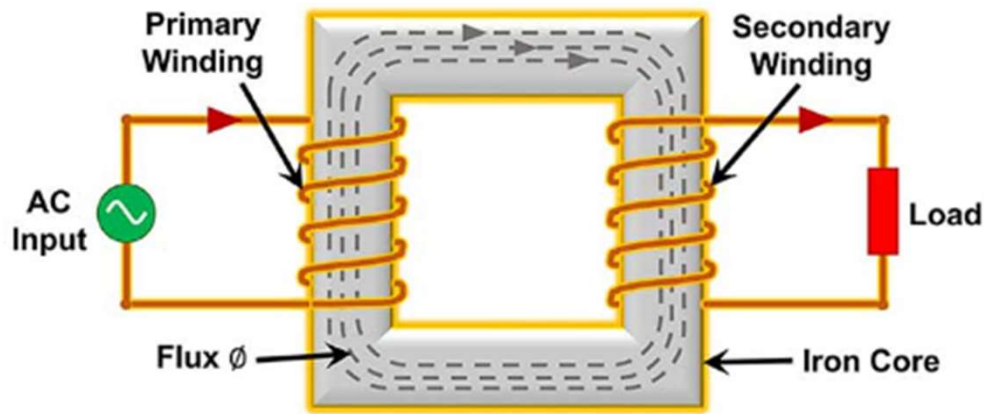
Transformer is an AC machine that

- transfers electrical energy from one electric circuit to another
- Does so without a change in frequency
- Does so by the principle of electromagnetic induction
- Has electric circuit that are linked by a magnetic circuit

Transformers require no moving parts , hence called as static transformer

- Due to which it has minimum maintenance
- No losses due to moving parts (ie no friction and windage losses)
- Have the maximum efficiency
 - typical transformer efficiencies at full load lie between 96% to 97%
 - Extremely large transformers have efficiencies upto 99%

Major components



Transformer has three main components :

1. Primary Winding

2. Magnetic Core

3. Secondary Winding

1) Primary Winding

- Transformer's primary windings when coupled to an electrical source, results in the production of magnetic flux.

2) Magnetic Core

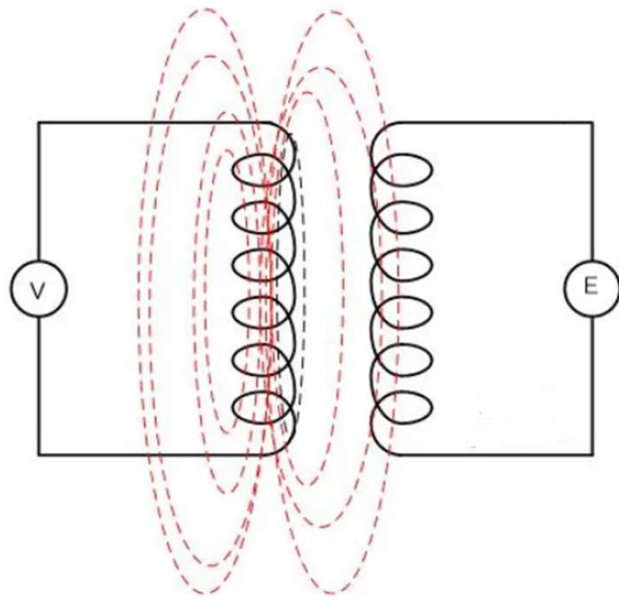
- The magnetic flux that is produced by the main winding, will flow through this low-reluctance path linked to the secondary winding, which will result in the formation of a closed magnetic circuit.

3) Secondary Winding

- When the flux that is created by the primary winding passes through the core, it will eventually connect up with the secondary winding.
- This winding generally winds on the same core as the other windings, and it provides the output that is initiated from the transformer.

Working Principle of a Transformer

- The basic principle behind working of a transformer is the phenomenon of mutual induction between two windings linked by common magnetic flux.
- The transformer is based on two principles:
 1. An electric current can produce a magnetic field (electromagnetism)
 2. A changing magnetic field within a coil of wire induces a voltage across the ends of the coil (electromagnetic induction). Changing the current in the primary coil changes the magnetic flux that is developed. The changing magnetic flux induces a voltage in the secondary coil.



- An alternating voltage source is connected with the primary winding which will produce a varying magnetic field when a current i flows through it. Now when we bring the secondary winding near to the primary, the changing magnetic field will link to it and according to the Faraday's Laws of electromagnetic induction an emf E will induced in the secondary side
- It is seen that the magnetic field lines generated due to the primary coil are not able to link completely with the secondary coil, so the losses are more. Thus a common magnetic medium is provided to link both the coils in a effective way.
- The material used for core building must be magnetic material with high relative permeability to give low reluctance path for magnetic flux, so it can collect most of the flux produced by the primary winding
- The magnetic core is made up of thin sheets called laminations of high grade steel silicon steel . The laminations reduce eddy current losses and silicon steel reduces hysteresis loss.

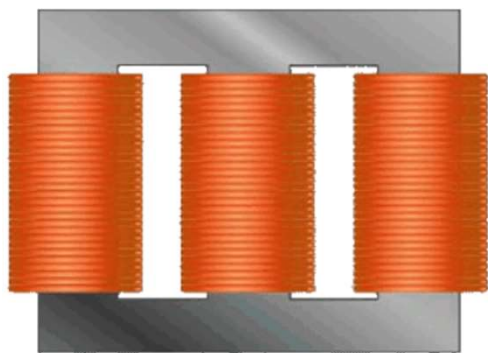
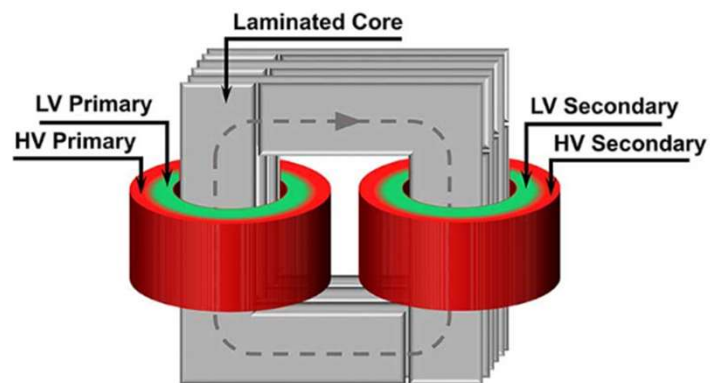
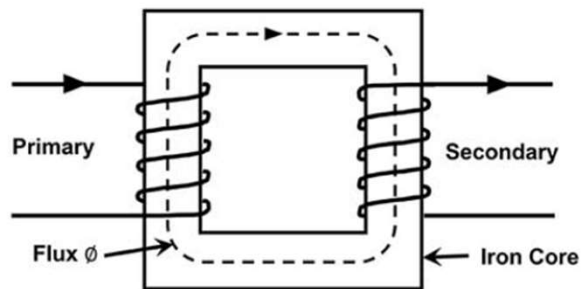
Working of a Transformer

- The primary and secondary coils of a transformer are electrically separated but magnetically linked to each other.
- When, primary winding is connected to a source of alternating voltage, alternating magnetic flux is produced around the winding
- The core provides magnetic path for the flux, to get linked with the secondary winding. Most of the flux gets linked with the secondary winding which is called as 'useful flux' or main 'flux', and the flux which does not get linked with secondary winding is called as 'leakage flux'.
- As the flux produced is alternating EMF (the direction of it is continuously changing), gets induced in the secondary winding. This emf is called 'mutually induced emf', and the frequency of mutually induced emf is same as that of supplied emf.
- If the secondary winding is closed circuit, then mutually induced current flows through it, and hence the electrical energy is transferred from one circuit (primary) to another circuit (secondary).
- The flux of the iron core links with the primary coil also , hence produced self induced emf in the primary winding. This self induced primary emf opposes the applied voltage (sometimes called Back emf of primary)

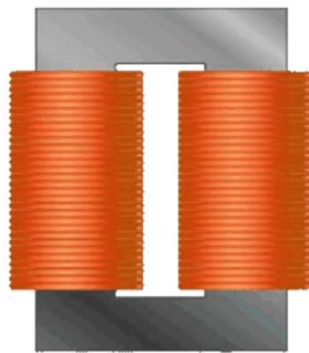
Classification of Transformer

1. Types of transformer based on the transformer's core	• Core Type Transformer • Shell Type Transformer
2. Types of transformer based on voltage conversion	• Step Up Transformer • Step Down Transformer
3. Types of transformer based on its purpose	• Power Transformer • Distribution Transformer • Isolation Transformer • Current Transformer • Potential Transformer
4. Transformer types based on windings	• Two Winding Transformer • Autotransformer
5. Transformer types based on the insulation used	• Dry Type Transformer • Oil Immersed Transformer
6. Transformer types based on the number of phases	• Single Phase Transformer • Three Phase Transformer

Core type transformer

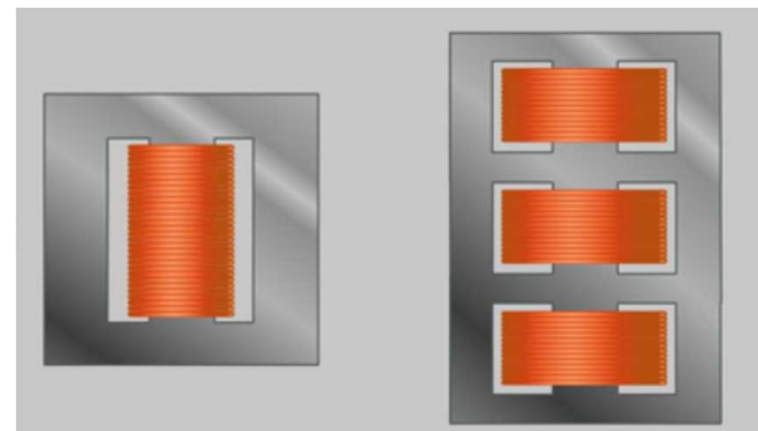
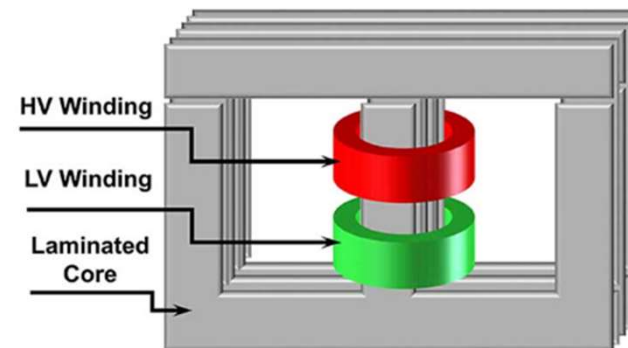
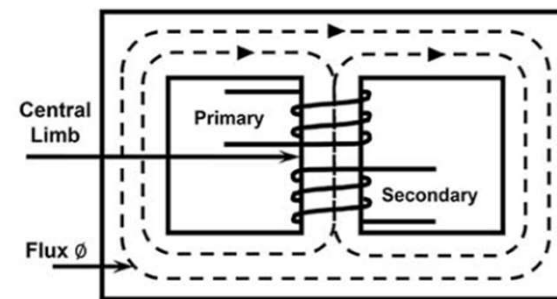


Three phase



Single phase

Shell type transformer



Ideal Transformer

- No winding resistance
- No magnetic leakage
- No iron loss
- Zero magnetising current

